第八章 Bose统计和Fermi统计.

§1 热力学量的统计表达式

$$a_l = \frac{\omega_l}{e^{\alpha+\beta\varepsilon_l} \mp 1}$$

引入巨配分函数 $\Xi = \prod_l (1 \mp e^{-\alpha-\beta\varepsilon_l})^{\mp\omega_l}$

$$\bar{N} = \sum a_l = -\frac{\partial}{\partial\alpha} \ln \Xi$$

$$U = \sum a_l \varepsilon_l = \sum \frac{\varepsilon_l \omega_l}{e^{\alpha+\beta\varepsilon_l} \mp 1} = -\frac{\partial}{\partial\beta} \ln \Xi$$

$$Y = \sum \frac{\partial \varepsilon_l}{\partial y} a_l$$

$$\frac{\partial}{\partial y} \ln \Xi = -\beta \sum \omega_l \ln(1 \mp e^{-\alpha-\beta\varepsilon_l}) \frac{\partial \varepsilon_l}{\partial y}$$

$$= \sum \frac{\omega_l}{e^{\alpha+\beta\varepsilon_l} \mp 1} \cdot \frac{\partial \varepsilon_l}{\partial y} = -\frac{1}{\beta} \frac{\partial}{\partial y} \ln \Xi$$

$$\Rightarrow d(\ln \Xi) = \frac{\partial \ln \Xi}{\partial \alpha} d\alpha + \frac{\partial \ln \Xi}{\partial \beta} d\beta + \frac{\partial \ln \Xi}{\partial y} dy$$

$$= -\bar{N} d\alpha - U d\beta - \beta Y dy.$$

$$\beta(dU - Y dy) = d(\beta U) - U d\beta + [d(\ln \Xi) + \bar{N} d\alpha + U d\beta]$$

$$= d(\ln \Xi + \beta U + \bar{N}\alpha) - \alpha d\bar{N}$$

$$= d\left(\ln \Xi - \beta \frac{\partial}{\partial \beta} \ln \Xi - \alpha \frac{\partial}{\partial \alpha} \ln \Xi\right) - \alpha d\bar{N}$$

$$\Rightarrow \beta\left(dU - Y dy + \frac{\alpha}{\beta} d\bar{N}\right) = d\left(\ln \Xi - \alpha \frac{\partial}{\partial \alpha} \ln \Xi - \beta \frac{\partial}{\partial \beta} \ln \Xi\right)$$

$$\frac{1}{T}(dU - Y dy - \mu d\bar{N}) = dS \quad \left(\alpha = -\frac{\mu}{kT},\ \beta = \frac{1}{kT}\right)$$

$$\Rightarrow S = k\left(\ln \Xi - \alpha \frac{\partial}{\partial \alpha} \ln \Xi - \beta \frac{\partial}{\partial \beta} \ln \Xi\right)$$

$$= k \sum \left\{\mp \omega_l \ln(1 \mp e^{-\alpha-\beta\varepsilon_l}) \pm \alpha \cdot \omega_l \cdot \frac{\pm e^{-\alpha-\beta\varepsilon_l}}{1 \mp e^{-\alpha-\beta\varepsilon_l}} + \beta \varepsilon_l \omega_l \frac{\pm e^{-\alpha-\beta\varepsilon_l}}{1 \mp e^{-\alpha-\beta\varepsilon_l}} \right]$$

$$= k \sum \left\{ \mp \omega_l \left[\ln\left(\frac{\omega_l}{a_l}\right) - (\alpha + \beta\varepsilon_l)\right] + (\alpha + \beta\varepsilon_l) a_l \right\}$$

$$= k \sum \left\{ \mp \omega_l \left[\ln\left(\frac{\omega_l}{a_l}\right) - \ln\left(\frac{\omega_l}{a_l} \pm 1\right)\right] + \ln\left(\frac{\omega_l}{a_l} \pm 1\right) a_l \right\}$$

$$= k \sum \left\{ \pm (\omega_l \pm a_l) \ln(\omega_l \pm a_l) \mp \omega_l \ln \omega_l \mp a_l \ln a_l \right\}$$

$$= k \ln \Omega$$

巨热力势 $J = U - TS - \mu N$

$$= -\frac{\partial}{\partial\beta}\ln\Xi - kT\left(\ln\Xi - \alpha\frac{\partial}{\partial\alpha}\ln\Xi - \beta\frac{\partial}{\partial\beta}\ln\Xi\right) + \frac{\alpha}{\beta}\frac{\partial}{\partial\alpha}\ln\Xi$$

$$= -kT\ln\Xi$$

§2 弱简并的 Bose / Fermi 气体.

$$\varepsilon = \frac{1}{2m}(p_x^2 + p_y^2 + p_z^2)$$

在体积 V 内，分布在 $(p \to p+dp)$ $\varepsilon \sim \varepsilon + d\varepsilon$ 的状态数.

$$D(p)\,dp = g\cdot\frac{V}{h^3}\cdot 4\pi p^2\,dp$$

$$D(\varepsilon)\,d\varepsilon = g\cdot\frac{V}{h^3}\,4\pi\cdot 2m\varepsilon\cdot\sqrt{\frac{m}{2}}\,\frac{1}{\sqrt{\varepsilon}}\,d\varepsilon$$

$$= g\cdot\frac{2\pi V}{h^3}(2m)^{3/2}\,\varepsilon^{1/2}\,d\varepsilon$$

对于能级 ε，平均占据数 对 Fermion，$n = 0 / 1$，对 Boson，$n = 0, 1, \cdots\cdots$

$$\langle n(\varepsilon)\rangle = \frac{\sum n e^{-n(\beta\varepsilon-\mu)}}{\sum e^{-n(\beta\varepsilon-\mu)}} = \begin{cases}\dfrac{0 + e^{-(\beta\varepsilon-\mu)}}{1 + e^{-(\beta\varepsilon-\mu)}} = \dfrac{1}{e^{\alpha+\beta\varepsilon}+1} & (\text{Fermion}) \\[2ex] \dfrac{e^{-\alpha-\beta\varepsilon}}{(1 - e^{-\alpha-\beta\varepsilon})} = \dfrac{1}{e^{\alpha+\beta\varepsilon}-1} & (\text{Boson})\end{cases}$$

$$\sum_{n=0}^{\infty} n x^n = \frac{x}{(1-x)^2}$$

$$N = \int D(\varepsilon)\langle n(\varepsilon)\rangle\,d\varepsilon = g\,\frac{2\pi V}{h^3}(2m)^{3/2}\int\frac{\varepsilon^{1/2}\,d\varepsilon}{e^{\alpha+\beta\varepsilon}\pm 1}$$

$$U = \int \varepsilon D(\varepsilon)\langle n(\varepsilon)\rangle\,d\varepsilon = g\,\frac{2\pi V}{h^3}(2m)^{3/2}\int\frac{\varepsilon^{3/2}\,d\varepsilon}{e^{\alpha+\beta\varepsilon}\pm 1}$$

记 $x = \beta\varepsilon$，在弱简并情形下，$e^{-\alpha} \ll 1$

$$N = g\cdot\frac{2\pi V}{h^3}(2mkT)^{3/2}\int\frac{x^{1/2}\,dx}{e^{\alpha+x}\pm 1}$$

$$\approx g\cdot\frac{2\pi V}{h^3}(2mkT)^{3/2}\int x^{1/2}e^{-(\alpha+x)}\left[1 \mp e^{-(\alpha+x)}\right]dx$$

$$= g\cdot\left(\frac{2\pi mkT}{h^2}\right)^{3/2} V e^{-\alpha}\left(1 \mp \frac{1}{2^{3/2}}e^{-\alpha}\right)$$

$$U = g \cdot \frac{2\pi V}{h^3} (2mkT)^{3/2} kT \cdot \int \frac{x^{3/2}\,dx}{e^{\alpha+x} \pm 1}$$

$$\approx g \cdot \frac{2\pi V}{h^3} (2mkT)^{3/2} kT \cdot \int x^{3/2} e^{-(\alpha+x)} \left[1 \mp e^{-(\alpha+x)}\right] dx$$

$$= \frac{3}{2} g \left(\frac{2\pi mkT}{h^2}\right)^{3/2} VkT e^{-\alpha} \left(1 \mp \frac{1}{2^{5/2}} e^{-\alpha}\right)$$

$$\Rightarrow U = \frac{3}{2} NkT \left(1 \pm \frac{1}{4\sqrt{2}} e^{-\alpha}\right)$$

$e^{-\alpha}$ 代入0级近似（Boltzmann分布）

$$e^{-\alpha} = \frac{N}{V} \left(\frac{h^3}{2\pi mkT}\right)^{3/2} \frac{1}{g}.$$

$$\approx \frac{3}{2} NkT \left[1 \pm \frac{1}{4\sqrt{2}} \frac{1}{g} \frac{N}{V} \left(\frac{h^2}{2\pi mkT}\right)^{3/2}\right]$$

$$= \frac{3}{2} NkT \left(1 \pm \underline{\frac{1}{4\sqrt{2}g} n\lambda^3}\right)$$

附加内能：Fermion > 0（排斥），Boson < 0（吸引）

§ 3 BEC

$n\lambda^3$ 较大时出现的反常现象.

$$a_l = \frac{\omega_l}{e^{\frac{\varepsilon_l-\mu}{kT}} - 1} > 0 \rightarrow \varepsilon_0 = 0 > \mu$$

μ 的约束 $n = \frac{N}{V} = \frac{2\pi}{h^3}(2m)^{3/2} \int_0^\infty \frac{\varepsilon^{1/2}\,d\varepsilon}{e^{\frac{\varepsilon-\mu}{kT}} - 1}$，$T \searrow \mu \nearrow$

在临界 $T = T_c$，$\mu \to 0^-$.

$$n = \frac{2\pi}{h^3}(2m)^{3/2} \int_0^\infty \frac{\varepsilon^{1/2}\,d\varepsilon}{e^{\varepsilon/kT_c} - 1}$$

$$= \frac{2\pi}{h^3}(2mkT_c)^{3/2} \underline{\int \frac{x^{1/2}\,dx}{e^x - 1}} = \frac{\sqrt{\pi}}{2} \times 2.612$$

$$\Rightarrow T_c = \frac{2\pi}{(2.612)^{2/3}} \frac{\hbar^2}{mk} n^{2/3}$$

$T < T_c$ 时，不能用 $n = \sum \rightarrow \int$ 近似，$\varepsilon = 0$ 的粒子数量多.

$$n = n_0(T) + \sum_{l=1}^{\infty} a_l = n_0(T) + \frac{2\pi}{h^3}(2m)^{3/2} \int \frac{\varepsilon^{1/2}\,d\varepsilon}{e^{\varepsilon/kT} - 1}$$

$$= n_0(T) + \frac{2\pi}{h^3}(2mkT_c)^{3/2} \int \frac{x^{1/2}\,dx}{e^x - 1}$$

$$= n_0(T) + n \cdot \left(\frac{T}{T_c}\right)^{3/2}$$

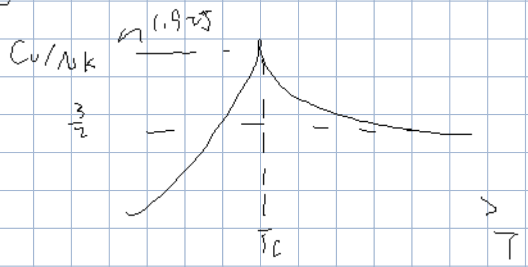
$\Rightarrow n_0(T) = n\left[1-\left(\frac{T}{T_c}\right)^{3/2}\right]$, BEC

$\varepsilon = 0$ 粒子对 u 无贡献.

$$u = \frac{2\pi V}{h^3}(2mkT)^{3/2}\,kT\cdot\int\frac{x^{3/2}\,dx}{e^x-1}$$

$$= 0.7\,NkT\left(\frac{T}{T_c}\right)^{3/2}$$

$$C_V = \frac{\partial u}{\partial T} = 1.925\,Nk\left(\frac{T}{T_c}\right)^{3/2}$$



§4 光子气体

$p = \hbar k$, $\varepsilon = cp$

光子服从玻色分布 $a_l = \frac{\omega_l}{e^{\beta\varepsilon_l}-1}$ （粒子数不守恒，$\alpha = 0$）

自旋 $= 1 \Rightarrow g = 2$

态密度, $D(p)\,dp = g\cdot\frac{4\pi V}{h^3}p^2\,dp = \frac{8\pi V}{h^3}p^2\,dp$

$D(\omega)\,d\omega = \frac{V}{\pi^2c^3}\omega^2\,d\omega$

平均光子数 $dN_\omega = D(\omega)\overset{=\frac{1}{e^{\beta\varepsilon}-1}}{\underline{\langle n(\omega)\rangle}}\,d\omega = \frac{1}{e^{\frac{\hbar\omega}{kT}}-1}\frac{V}{\pi^2c^3}\omega^2\,d\omega$

内能 $u(\omega,T)\,d\omega = \hbar\omega\cdot\langle n(\omega)\rangle\,d\omega$

$= \frac{V}{\pi^2c^3}\frac{1}{e^{\frac{\hbar\omega}{kT}}-1}\hbar\omega^3\,d\omega$ Planck公式

$$= \begin{cases}\frac{V}{\pi^2c^3}kT\omega^2\,d\omega & (\omega \ll \frac{kT}{\hbar}\text{，低频})\\ \frac{V}{\pi^2c^3}e^{-\frac{\hbar\omega}{kT}}\hbar\omega^3\,d\omega & (\omega \gg \frac{kT}{\hbar}\text{，高频})\end{cases}$$

$$u = \frac{V}{\pi^2c^3}\int_0^\infty\frac{\hbar\omega^3}{e^{\hbar\omega/kT}-1}\,d\omega = \frac{V\hbar}{\pi^2c^3}\left(\frac{kT}{\hbar}\right)^4\int_0^\infty\frac{x^3}{e^x-1}\,dx$$

$$= \frac{\pi^2k^4}{15c^3\hbar^3}VT^4$$

最概然频率: $\frac{d}{d\omega}u(\omega,T) = 0 \Rightarrow \omega_m = 2.82\cdot\frac{kT}{\hbar}$

巨配分函数 $\ln \Xi = -\sum \omega_i \ln(1-e^{-\alpha-\beta\varepsilon_i})$

$$= -\int_0^\infty \frac{V}{\pi^2 c^3}\omega^2 \ln(1-e^{-\beta\hbar\omega})\,d\omega$$

$$= -\frac{V}{\pi^2 c^3}\left(\frac{kT}{\hbar}\right)^3 \int_0^\infty x^2 \ln(1-e^{-x})\,dx$$

$$= -\frac{V}{\pi^2 c^3}\left(\frac{kT}{\hbar}\right)^3 \left[\frac{1}{3}x^3 \ln(1-e^{-x})\Big|_0^\infty - \int_0^\infty \frac{1}{3}x^3 \frac{e^{-x}}{1-e^{-x}}\,dx\right]$$

$$= \frac{V}{3\pi^2 c^3}\left(\frac{kT}{\hbar}\right)^3 \int_0^\infty \frac{1}{e^x-1}x^3\,dx$$

$$= \frac{V}{45\pi^2 c^3}\left(\frac{kT}{\hbar}\right)^3$$

$u = -\frac{\partial}{\partial\beta}\ln\Xi = \frac{VkT}{15\pi^2 c^3}\left(\frac{kT}{\hbar}\right)^3 \propto T^4$ S-B定律

$p = \frac{1}{\beta}\frac{\partial}{\partial V}\ln\Xi = \frac{kT}{45\pi^2 c^3}\left(\frac{kT}{\hbar}\right)^3 \Rightarrow pV = \frac{1}{3}u$

$S = k\left(\ln\Xi - \beta\frac{\partial}{\partial\beta}\ln\Xi\right)$

$$= \frac{4kV}{45\pi^2 c^3}\left(\frac{kT}{\hbar}\right)^3$$

§5. 强简并理想Fermi气体 - 金属中的自由电子.

平均电子数 $f = \frac{1}{e^{\frac{(\varepsilon-\mu)}{kT}}+1}$

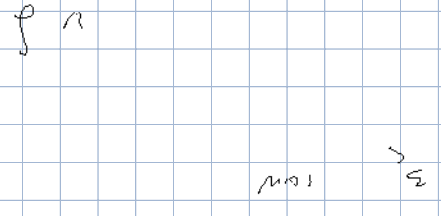
态密度 $D(p)\,dp = g\cdot\frac{V}{h^3}4\pi p^2\,dp = \frac{8\pi V}{h^3}p^2\,dp$

$$D(\varepsilon)\,d\varepsilon = \frac{4\pi V}{h^3}(2m)^{3/2}\varepsilon^{1/2}\,d\varepsilon$$

$dN_\varepsilon = f\cdot D(\varepsilon)\,d\varepsilon$

$$= \frac{4\pi V}{h^3}(2m)^{3/2}\cdot\frac{\varepsilon^{1/2}\,d\varepsilon}{e^{\frac{\varepsilon-\mu}{kT}}+1}$$

(1) $T=0 \Rightarrow f = \begin{cases} 1 & \varepsilon<\mu(0) \\ 0 & \varepsilon>\mu(0)\end{cases}$



约束 $\frac{4\pi V}{h^3}(2m)^{3/2}\int^{\mu(0)}\varepsilon^{1/2}\,d\varepsilon =$

h^3 0 N

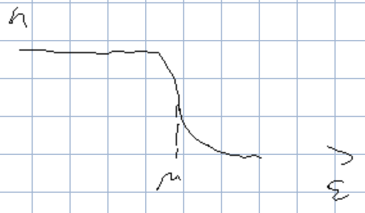
解得 $\mu(0) = \frac{\hbar^2}{2m}\left(3\pi^2 \frac{N}{V}\right)^{2/3} = \varepsilon_F$

定义 Fermi 动量 $p_F = \sqrt{2m\varepsilon_F} = \hbar\left(3\pi^2\frac{N}{V}\right)^{1/3}$

内能 $U(0) = \frac{4\pi V}{h^3}(2m)^{3/2}\int_0^{\varepsilon_F} \varepsilon^{3/2}\, d\varepsilon$

$= \frac{3}{5} N \varepsilon_F$

(2) $T > 0$



只有能量在 μ 的 kT 邻域分布的电子，

才对热容有贡献：

$$N_{eff} \approx N \cdot \frac{kT}{\mu}$$

$$C_V^e = \frac{3}{2} N_{eff} k = \frac{3}{2} N k \frac{kT}{\mu} = \frac{3}{2} N k \underline{\frac{T}{T_F}} \ll 1$$

$$N = \frac{4\pi V}{h^3}(2m)^{3/2}\int_0^\infty \frac{\varepsilon^{1/2}\, d\varepsilon}{e^{\frac{\varepsilon-\mu}{kT}}+1} \to \text{可解出 } \mu$$

$$U = \frac{4\pi V}{h^3}(2m)^{3/2}\int_0^\infty \frac{\varepsilon^{3/2}\, d\varepsilon}{e^{\frac{\varepsilon-\mu}{kT}}+1}$$

计算 $I = \int_0^\infty \frac{\eta(\varepsilon)\, d\varepsilon}{e^{\frac{\varepsilon-\mu}{kT}}+1}$ ...

$N = \frac{2}{3} C \mu^{3/2}\left[1 + \frac{\pi^2}{8}\left(\frac{kT}{\mu}\right)^2\right]$ 解得 $\mu = \left(\frac{3N}{2C}\right)^{2/3}\left(1 + \frac{\pi^2}{8}\left(\frac{kT}{\mu}\right)^2\right)^{-2/3}$

$$U = \frac{2}{5} C \mu^{3/2}\left[1 + \frac{5\pi^2}{8}\left(\frac{kT}{\mu}\right)^2\right]$$

$$C = \frac{4\pi V}{h^3}(2m)^{3/2}$$

$$\mu \approx \mu(0)\left[1 + \frac{\pi^2}{8}\left(\frac{kT}{\mu}\right)^2\right]^{-2/3} \approx \mu(0)\left\{1 - \frac{\pi^2}{12}\left[\frac{kT}{\mu(0)}\right]^2\right\}$$

$$U \approx \frac{2}{5} C \cdot \mu(0)^{5/2}\left\{1 - \frac{\pi^2}{12}\left[\frac{kT}{\mu(0)}\right]^2\right\}^{5/2}\left\{1 + \frac{5}{8}\pi^2\left[\frac{kT}{\mu(0)}\right]^2\right\}$$

$$= \frac{3}{5} C \mu(0)^{5/2}\left\{1 + \frac{5}{12}\left[\frac{kT}{\mu(0)}\right]^2\right\}$$

$$C_V = \left(\frac{\partial U}{\partial T}\right)_V = Nk\frac{\pi^2}{2}\frac{kT}{\mu(0)} = \gamma_0 T \quad \propto T$$